

Assignment 2

(Marks = 100)

Important Points:

- Show complete working for each question.
- Submit handwritten copy for Section A and Section C, while submit .py(.ipynb) file for Section B.
- Plagiarism will lead to negative marking.

Section A: Genetic Algorithm

Question 1: (Marks = 20)

Maximize:

$$f(x) = x^2 \quad \text{where } x \in [0, 31]$$

Where:

Population size is 4

GP = 0.5

Encoding= Not Specified (Student must decide)

Crossover: Single-point crossover (Crossover point = 3)

Mutation: Bit-flip mutation (mutation rate = 0.1)

Selection: Roulette Wheel Selection

Tasks:

- 1) Define suitable chromosome representation for x.
- 2) Generate an initial population (randomly assume values).
- 3) Decode chromosomes and compute fitness values.

- 4) Perform rank based selection (show probabilities & range)(Assume random numbers $r_1=0.35$ and $r_2=0.60$)
- 5) Select parents based on generation gap.
- 6) Apply crossover and show offspring.
- 7) Apply mutation and show final population.

Question 2: (Marks = 30)

You are a logistics engineer for SkySwift, a boutique travel agency. You need to find the most efficient flight sequence for a traveler visiting four cities in a single loop: A (Start/End), B, C, and D. The traveler must visit every city exactly once and return to City A.

Your goal is to minimize the **Total Travel Cost_{total}**, which is a weighted combination of ticket prices and travel time.

Use the following table for your calculations:

Route	Price (\$)	Time (hrs)	Route	Price (\$)	Time (hrs)
A → B	200	2	B → D	250	4
A → C	450	5	C → B	150	2
A → D	300	3	C → D	300	3
B → A	200	2	D → A	300	3
B → C	150	2	D → C	300	3

Fitness function is given below:

$$C_{total} = \sum(\text{Ticket Price}) + [50 \times \sum(\text{Travel Time in hours})]$$

$$F = \frac{10,000}{C_{total}}$$

Apply genetic algorithms to find optimal path which starts and ends at A.

- a) Find an appropriate chromosome representation for population size 4.
- b) Apply fitness function
- c) Apply roulette wheel selection where generation gap is 0.5 (Select same as above points).
- d) Use single point crossover at the first index (after the first gene)
- e) Select an appropriate mutation mechanism i.e Swap/Scramble.

Repeat the algorithm for first two generations.

Section B: Hill Climbing Algorithm (Marks = 20)

Write the python code for solving the 8 Queens Problem using Hill Climbing Algorithm. Read 4.1.1 of the textbook (Stuart Russel. 3rd edition) particularly Figure 4.3 to understand heuristic cost function.

Section C: Linear Regression (Marks = 30)

A researcher is studying the relationship between hours studied (X) and exam score (Y) for a group of students.

Student	Hours Studied (X)	Score (Y)
1	1	2
2	2	4
3	3	5
4	4	4
5	5	5

- Find Slope β_1 and Intercept β_0 .
- Calculate Coefficient of Determination (R^2).
- Using your regression model, predict the score when: X=6 hours studied